

# Explainable Multi-Stage Deep Learning for Brain Tumor Segmentation and Classification

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## 1. Introduction

Brain tumors exhibit substantial variability in malignancy and morphology, which contributes to high rates of misdiagnosis with standard radiological assessment and thereby delaying critical interventions in neuro-oncology. Accurate and timely diagnosis of brain tumors therefore remains a critical challenge in neuro-oncology. Conventional MRI interpretation by radiologists yields misdiagnosis rates exceeding 20% because of the broad variability in tumor appearance, anatomical location, and degree of malignancy [1]. Such diagnostic inaccuracies defer life-saving therapies and increase patient morbidity. Recent advances in deep learning have shown promise for automated tumor classification; however, most existing models remain black-box systems that lack the level of interpretability required for clinical adoption.

We therefore propose an explainable multi-stage deep learning framework that jointly performs segmentation, classification, and report generation. The pipeline (1) isolates tumor regions with a U-Net segmentation network, (2) classifies the segmented patches using a transfer-learned ResNet50 CNN [2], and (3) produces visual heatmaps via Grad-CAM together with natural-language explanations via a fine-tuned CLIP-based vision-language model [3]. By enforcing an explicit segmentation stage, we reduce background noise and improve both accuracy and explainability. All components are built upon publicly available datasets and open-source implementations, with modest modifications for medical-report style and class-imbalance handling. Our work bridges the gap between high-performing vision models and the transparency required for clinical adoption.

Using the public Kaggle Brain Tumor MRI Dataset (four classes: glioma, meningioma, pituitary tumor, and no tumor) and the SciDB Brain Tumor Dataset with pixel-level masks, our preliminary results achieve a Dice Similarity Coefficient of 0.7502 for segmentation. Classification

training using the connected pipeline is currently underway. Qualitative evaluation confirms that the segmentation outputs align closely with ground-truth masks and that the overall framework is well positioned for clinical-style interpretability.

The overall plan is to develop this integrated, clinically relevant system through a modular pipeline that progresses from data preparation and baseline training (completed at this milestone) to full VLM fine-tuning, end-to-end integration, hyperparameter optimization, and rigorous clinical-style validation in later phases.

## 2. Problem Statement

The precise problem is to automatically classify brain tumors into one of four categories—"glioma", "meningioma", "pituitary tumor", or "no tumor" from 2D MRI slices while simultaneously generating an explanatory report that improves clinical trust and interpretability. The primary dataset is the publicly available Brain Tumor MRI Dataset hosted on Kaggle, consisting of approximately 7,000 human brain MRI images distributed across the four classes (no new data collection is required). Supplementary segmentation annotations are drawn from the SciDB Brain Tumor Dataset, which provides pixel-level masks for tumor regions.

Expected results include: (1) a segmentation model that accurately isolates tumor boundaries, (2) a classifier achieving high classification performance (target: greater than 95% accuracy, precision, recall, and F1-score), (3) Grad-CAM heatmaps that focus on true tumor regions rather than background, and (4) concise, medically coherent reports generated by a VLM that describe tumor characteristics, confidence, and rationale.

Evaluation will be both quantitative and qualitative. For segmentation: Dice Similarity Coefficient, Hausdorff Dis-

tance, sensitivity, and specificity. For classification: accuracy, precision, recall, F1-score, and confusion matrices. Qualitative assessment will involve visual inspection of segmentation overlays against ground-truth masks, Grad-CAM heatmaps, and the quality of human-readable reports. All experiments will be conducted with stratified train/validation/test splits (80/10/10) and appropriate handling of class imbalance.

### 3. Technical Approach

Our pipeline consists of three modular stages (see Figure 1).

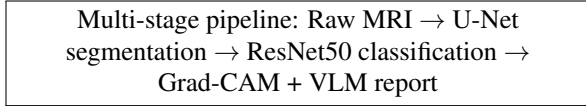


Figure 1. Overview of the proposed explainable multi-stage framework.

#### 3.1. Preprocessing & Segmentation Stage

Input MRI slices are passed through a U-Net segmentation model. While we initially experimented with nnU-Net for its self-configuring architecture and state-of-the-art performance on medical segmentation tasks, we decided to adopt a standard U-Net architecture because training nnU-Net on an A100 GPU in Colab for 50 epochs already takes approximately 1 hour, while achieving good results with nnU-Net typically requires around 1000 epochs. This makes U-Net a more practical and computationally feasible choice for our resources and project timeline while still delivering strong segmentation performance. This step isolates tumor regions, removes irrelevant background noise, and crops and resizes the region of interest to focus subsequent classification on diagnostically relevant features. Segmentation masks from the SciDB dataset serve as ground truth during training (see Figure 2).

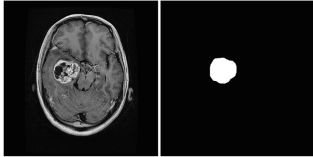


Figure 2. Segmented mask samples from SciDB dataset.

#### 3.2. Classification Stage

The segmented tumor regions are fed into a fine-tuned CNN classifier. ResNet50 (pre-trained on ImageNet) [2] is employed via transfer learning, with the final fully connected layer replaced by a 4-class softmax output. Hyperparameters (learning rate, batch size, epochs) are

optimized using grid search; data augmentation (rotation, flipping, brightness/contrast jitter) and class-weight balancing address dataset imbalances.

#### 3.3. Explainability & Report Generation

After classification, Grad-CAM generates visual heatmaps highlighting the regions that most influenced the model’s decision. These heatmaps are overlaid on the original MRI for qualitative validation. A vision-language model (based on the CLIP architecture) [3] is fine-tuned on medical report templates to produce natural-language outputs describing tumor type, morphological characteristics, confidence score, and clinical rationale.

### 4. Preliminary Results

We have completed data preprocessing and U-Net segmentation training for 90 epochs and are currently working on the segmentation-to-classification connection and subsequent classification fine-tuning.

#### 4.1. Data Preparation

Fully completed. The Kaggle Brain Tumor MRI Dataset has been downloaded, cleaned, and split into training, validation, and test sets using an 80/10/10 stratified split. Segmentation masks from the SciDB Brain Tumor Dataset have been successfully aligned with the corresponding MRI slices. Exploratory data analysis was performed to understand class distribution and image characteristics.

#### 4.2. Segmentation Module

Following our decision to switch from nnU-Net due to computational constraints where nnU-Net training on an A100 GPU already takes approximately 1 hour for 50 epochs, while the default training schedule requires around 1000 epochs to achieve good results [4], the U-Net segmentation model has been successfully trained for 90 epochs. The segmentation model achieves the following performance on the validation set:

- Dice Similarity Coefficient: 0.7502
- Sensitivity: 0.7666
- Specificity: 0.9256

Visual evaluation of segmentation overlays confirms reasonable agreement with ground-truth tumor boundaries, with some discrepancies observed in low-contrast and small tumor regions.

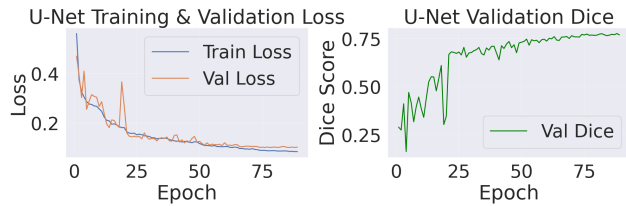


Figure 3. U-Net training and validation loss curves (left) and validation Dice score (right) over 90 epochs.

### 4.3. Segmentation-to-Classification Connection

We have completed the segmentation stage and are currently implementing and testing the connection between the U-Net output and the classification module. This includes post-processing of U-Net masks (thresholding, connected-component analysis, and tumor-region cropping) to generate clean, properly formatted patches as input for the ResNet50 classifier.

### 4.4. Classification Module

Training of the fine-tuned ResNet50 classifier on the segmented tumor regions has not yet commenced. The model will be trained using transfer learning with data augmentation and class-balancing techniques. Full evaluation metrics (accuracy, precision, recall, F1-score, and confusion matrix) will be reported once training is completed in the next phase.

### 4.5. Explainability & Report Generation

The Grad-CAM module has been set up and is ready for heatmap generation once classification training begins. Preliminary setup for the CLIP-based vision-language model (template-based prompting) has also been completed and will be fully integrated after classification results are obtained.

## References

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